HealthConnect Clinic Experience Lab

Week 4 – Initial Analysis Document

AnalystLab Africa Experience Lab Internship Programme
Track: Data Analytics
Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

1. Project Overview

HealthConnect Clinic is a fictional healthcare provider experiencing operational and patient-support challenges related to missed appointments, inefficient use of appointment slots, and repetitive appointment enquiries. The central project question is: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

As a Data Analytics intern, my Week 4 focus is to understand the appointment data and identify how it can be used to investigate appointment attendance and no-show patterns. This document establishes the analytical foundation for the subsequent stages of the HealthConnect project.

2. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 appointment records and 18 variables. The dataset is fictional and anonymized and includes information about patient demographics, appointment details, booking information, previous appointment history, reminders, distance to the clinic, waiting time, and appointment outcomes.

Category	Variables	Purpose
Identification	appointment_id, patient_id	Identify appointments and patients
Patient information	gender, age, age_group	Describe patient characteristics
Appointment information	appointment_type, appointment_day, appointment_time	Describe appointment characteristics
Booking information	booking_date, appointment_date, booking_lead_days	Assess scheduling and lead time
Previous history	previous_appointments, previous_no_shows	Assess attendance history
Reminder information	reminder_sent, reminder_channel	Assess reminder engagement
Access / experience	distance_to_clinic_km, waiting_time_minutes	Assess access and service experience
Outcome	appointment_outcome	Identify attendance outcome

3. Data Quality Assessment

An initial assessment of the dataset indicates that it is suitable for further analysis. The review found no completely duplicated appointment records. The Data Dictionary indicates that distance_to_clinic_km and waiting_time_minutes contain limited missing values, so these fields will require attention during data preparation. Appointment IDs are unique, while patient IDs may repeat because a patient can have multiple appointments.

The main categorical variables contain consistent values, and the available numerical, date, and categorical fields are appropriate for analytical work. The primary outcome variable is appointment_outcome, which distinguishes Attended, No-Show, and Cancelled outcomes.

The assessment also identified important analytical variables including previous_no_shows, reminder_sent, reminder_channel, booking_lead_days, appointment_type, age_group, appointment_day, appointment_time, distance_to_clinic_km, and waiting_time_minutes.

4. Relevant Business Questions

1. What factors are associated with patients missing their scheduled appointments?
2. Does the number of previous no-shows relate to a patient's likelihood of missing their current appointment?
3. Does sending appointment reminders affect appointment attendance?
4. Does the time between booking and the appointment influence the likelihood of a no-show?
5. Are no-show patterns different across appointment types, patient age groups, appointment days, or appointment times?
6. Does distance to the clinic have an association with appointment attendance?
7. Is waiting time associated with appointment outcomes?

5. Potential KPIs

Potential KPI	Linked Business Question	Relevance
Appointment No-Show Rate	What factors are associated with missed appointments?	Provides an overall measure of the missed-appointment problem and a primary outcome measure.
Reminder Effectiveness Rate	Does sending appointment reminders affect attendance?	Helps assess whether reminder activity is associated with better attendance.
Previous No-Show Rate	Does previous no-show history relate to current attendance?	Helps identify whether previous missed appointments are associated with future no-shows.
Average Booking Lead Time	Does booking lead time influence no-shows?	Helps assess whether longer or shorter periods between booking and appointment are associated with missed appointments.
No-Show Rate by Appointment Type	Are no-show patterns different across appointment types?	Helps identify appointment categories that may require additional attention.

At Week 4, these KPIs are proposed and justified rather than calculated. Detailed KPI calculations and visualisations will be part of subsequent analysis stages.

6. Initial Analysis Approach

Data preparation and validation: Use a separate working copy for cleaning and transformation; confirm data types, check completeness and duplicates, review categorical consistency, and document significant changes.

Explore appointment outcomes: Examine appointment_outcome to establish the baseline distribution of Attended, No-Show, and Cancelled appointments.

Investigate factors associated with no-shows: Compare appointment outcomes with previous no-shows, reminders, booking lead time, appointment type, age group, appointment day/time, distance, and waiting time.

Analyse proposed KPIs: In later stages, calculate and monitor the five proposed KPIs and examine relevant breakdowns by patient and appointment characteristics.

Use appropriate analytics tools: Use SQL for data exploration and analytical queries and Power BI for later KPI analysis and visualisation, with data preparation performed as needed.

Identify actionable patterns: Focus on patterns that can inform appointment management and patient engagement while distinguishing association from causation.

7. Assumptions

- appointment_outcome accurately represents whether an appointment was attended, missed, or cancelled.
- Appointment and booking dates are recorded correctly.
- previous_no_shows represents the patient's recorded previous missed appointments.
- reminder_sent accurately indicates whether a reminder was sent.
- The available variables provide a useful starting point for identifying initial no-show patterns.

8. Limitations

- The dataset is fictional and anonymized, so findings cannot automatically be generalized to real healthcare facilities.
- The dataset may not contain every factor that influences whether a patient attends an appointment.
- An observed relationship between a variable and no-shows does not necessarily establish causation.
- Reminder data may indicate that a reminder was sent without showing whether it was received, read, or acted upon.
- Week 4 is an initial planning and understanding stage; final conclusions should not be drawn at this stage.

9. Risks and Dependencies

Risk / Dependency	Potential Impact	Mitigation / Requirement
Incorrect interpretation of variables	Misleading analysis	Use the Data Dictionary before detailed analysis.
Data quality issues discovered later	Affects KPI calculations	Validate data before detailed analysis.
Treating correlation as causation	Incorrect recommendations	Clearly distinguish association

		from causation.
Overlooking important variables	Incomplete findings	Review all relevant dataset fields.
Incorrect handling of cancellations	Distorts no-show analysis	Keep Cancelled and No-Show outcomes distinct.
Project resources	Analysis may be inconsistent	Use the dataset and Data Dictionary as core project resources and preserve original files.

10. Week 4 Project Summary

The specific Data Analytics problem is to investigate appointment attendance and no-show patterns at HealthConnect Clinic and establish an analytical foundation for reducing missed appointments and improving patient engagement.

The main resources used are the HealthConnect Appointment Dataset and the HealthConnect Data Dictionary. The dataset provides patient, appointment, booking, historical, reminder, access, waiting-time, and outcome information relevant to the problem.

The initial review indicates that the dataset is suitable for further analysis and contains several variables that can be examined in relation to appointment outcomes. Key areas include previous no-shows, reminders, booking lead time, appointment characteristics, distance, and waiting time.

The proposed approach is to prepare and validate the data, explore appointment outcomes, investigate factors associated with no-shows, and later calculate the proposed KPIs using appropriate analytical tools such as SQL and Power BI.

Key considerations include the fictional and anonymized nature of the dataset, possible unobserved factors, the distinction between cancellations and no-shows, and the need to avoid interpreting association as causation.

The proposed Week 5 focus is to move from problem understanding into practical data exploration and analysis, including KPI calculation, investigation of no-show patterns, comparisons across relevant variables, and development of evidence-based findings.

11. Conclusion

Week 4 establishes a structured foundation for the HealthConnect Data Analytics work. The business problem, relevant data, business questions, proposed KPIs, analytical approach, and key project considerations have been defined. This foundation will guide the detailed analysis and development activities in the next stage of the Experience Lab.

12. Submission Checklist

- Initial Analysis Document completed.
- Week 4 Project Summary included.
- Supporting HealthConnect project files organised separately from original resources.
- Google Drive folder organised and accessible to the assigned mentor.
- GitHub repository updated where applicable.

- README.md updated to reflect Week 4 progress.
- Final work reviewed for professional presentation and accuracy.